



Groundwater and Surface Water Quality for Irrigation in the Brazilian Semi-arid Region: Hydrochemical and Microbiological Assessment

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Abstract Anthropogenic pressures and climate change are increasing the risk of water quality degradation in semi-arid regions, particularly where water resources are intensively exploited for irrigation. This study evaluated the quality of groundwater (wells tapping sedimentary and crystalline aquifers) and surface water (the Jaguaribe River and the Orós Reservoir) in Ceará, Brazil, through physicochemical and microbiological analyses, irrigation suitability

indices, and multivariate statistical approaches. The results revealed pronounced seasonal variability. During the dry season, wells located in the crystalline basement (G8 and G9) exhibited electrical conductivity values exceeding 0.93 dS m^{-1} and total dissolved solids above 600 mg L^{-1} , characterizing brackish waters, together with elevated concentrations of iron (up to 5.74 mg L^{-1} in G4) and manganese ($> 1.6 \text{ mg L}^{-1}$ in G3). The highest pH value (9.6) was recorded at site R7, suggesting eutrophication-related processes. Although the Sodium Adsorption Ratio (SAR) remained low (< 3), the soluble sodium percentage

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exceeded 50% at R12. In comparison, the magnesium hazard index surpassed 50% at most sites during the dry season, indicating a potential risk of soil structural degradation. In the rainy season, dilution reduced salinity levels but increased microbiological contamination, with *Escherichia coli* detected in wells (G2 and G3) and river sites (R7 and R12), ranging from 9.3 to 15 CFU. In contrast, total coliform counts reached 46 MPN mL⁻¹. Principal Component Analysis distinguished the effects of lithology and evapotranspiration, with the first two components explaining 61.1% of the total variance. These findings provide valuable insights for water monitoring and sustainable management to enhance water and agricultural security in semi-arid regions.

Keywords Irrigation suitability indices · Jaguaribe River · Orós Reservoir · Salinity and sodicity risk · *Escherichia coli*

1 Introduction

Irrigation represents a strategic pillar for agricultural production and food sovereignty, even, in regions where low and irregular rainfall severely limits rain-fed crop cultivation. In semi-arid regions worldwide, irrigation enables multiple production cycles throughout the year, mitigates climatic risks, stabilizes yields, and increases farmers' income. Moreover, irrigation plays a crucial role in promoting the productive inclusion of smallholders and sustaining the export capacity of large-scale producers, thereby fostering regional socio-economic development and supporting rural livelihoods (Zhang et al., 2024). However, its use must be guided by rigorous technical criteria and the rational management of water resources, incorporating robust monitoring methodologies to assess the quality of irrigation water sources (Javadi et al., 2025; Macedo et al., 2024; Mitroi et al., 2025; Santos et al., 2025; Vale et al., 2024).

Assessing water quality for irrigation is critical in semi-arid environments, where water scarcity constitutes one of the main constraints to agricultural sustainability. Global projections indicate that, by 2050, water demand could increase by 50% to 55%, while by 2030, nearly half of the world's population may face severe water scarcity (Abd-Elhamid et al., 2021; Alharbi et al., 2024; He et al., 2021). In the Brazilian

semi-arid region, rainfall is projected to decrease by 10–20% by 2040, exacerbating water deficits, intensifying aridification and desertification processes, and increasing irrigation demand (Gonçalves & Dias, 2024; Marengo et al., 2018).

In this context, ensuring the quality of water sources used for irrigation is not only a technical requirement but also a strategic necessity to secure water resources and align agricultural systems with the United Nations 2030 Agenda for Sustainable Development Goals (SDG). Scientific evidence underscores that water security drives not only food security (SDG 2) and human health (SDG 3) but also strengthens water governance (SDG 6), responsible consumption and production (SDG 12), biodiversity conservation (SDG 15), and climate change mitigation (SDG 13) (Alum et al., 2024; Ishaque et al., 2024). According to climate projections from the Intergovernmental Panel on Climate Change (IPCC), there is a consistent trend toward more frequent and intense droughts across different parts of the globe, particularly in areas already vulnerable to water scarcity (IPCC, 2023).

In the northern portion of the Brazilian semi-arid region lies the state of Ceará, where approximately 92% of the territory is subject to arid conditions, extreme rainfall variability, prolonged and intense droughts, and high temperatures (Oliveira et al., 2025; Seigerman et al., 2024). Despite the adoption of water management strategies and hydraulic infrastructure, most municipalities in this state have not experienced significant reductions in climate vulnerability (Mitroi et al., 2025). One such municipality is Iguatu, where intensifying droughts have reduced water bodies and consequently increased irrigation demand to sustain agricultural production (Oliveira et al., 2025). Irrigation water demand in this area is met by groundwater, tributaries of the Jaguaribe River, and the Trussu and Orós Reservoirs, which are extensively exploited for local agriculture. Under these conditions, evaluating irrigation water quality becomes essential, as water scarcity imposes severe challenges to agricultural sustainability.

Specifically, for the Jaguaribe River Basin, where the present study was conducted, Gondim et al. (2018) projected that climate change may increase reference evapotranspiration by 2.3–6.3% and irrigation water requirements by 2.8–16.7% across different greenhouse gas emission scenarios. Most climate

models also predicted reductions in annual river discharge, indicating a progressive decline in future water availability. These projections are consistent with the findings of Oliveira et al. (2025), who demonstrated that the intensification of droughts in Iguatu over 37 years (1985–2023) resulted in a substantial reduction in surface water bodies and a decline in rainfed agricultural production, while simultaneously increasing reliance on irrigation to sustain agricultural activities. This vulnerability became particularly evident during the severe drought that affected northeastern Brazil between 2012 and 2018, when the Banabuiú Reservoir, one of the main strategic water sources supplying irrigated districts in Ceará, approached critical storage levels, highlighting the susceptibility of regional irrigation systems to prolonged drought events (Henriquez et al., 2026). In this context, assessing irrigation water quality is not merely a technical requirement but a strategic necessity, as the progressive salinization and contamination of increasingly limited water resources may substantially reduce the resilience of agricultural systems already subjected to growing hydroclimatic stress (Abd-Elhamid et al., 2021; Marengo et al., 2018).

The presence of soluble salts, potentially toxic elements, and compounds that alter pH and electrical conductivity directly compromises the soil's physicochemical structure, restricts nutrient uptake by plants, and interferes with their physiological development. Moreover, indices such as the Sodium Adsorption Ratio (SAR), Magnesium Adsorption Ratio (MAR), Magnesium Hazard (MH), Residual Sodium Carbonate (RSC), Soluble Sodium Percentage (SSP), Kelly's Index (KI), Permeability Index (PI), and total hardness can serve as effective indicators of water quality (Damiba et al., 2025; El Behairy et al., 2021; Mokhtar et al., 2022; Thapa et al., 2017). In this context, systematic monitoring of physicochemical parameters, water quality indices, and microbiological analyses becomes an indispensable strategy to ensure the rational and safe use of water resources, as well as to mitigate the risks of soil salinization and sodification.

Existing studies in the Brazilian semi-arid region have predominantly focused either on groundwater salinization during prolonged drought periods (Almeida et al., 2025; Rojas et al., 2020) or on isolated assessments of microbiological deterioration in surface waters (Bagordo et al., 2024; Silva et al., 2024). Although seasonal hydrogeochemical variability has

been investigated (Aditya et al., 2024) and the susceptibility of crystalline aquifers to salinization is well recognized (Egbueri et al., 2025; Teramoto et al., 2023), a substantial knowledge gap persists regarding the combined influence of hydrogeochemical evolution, trace metal mobilization, particularly Fe and Mn, and fecal indicator bacteria dynamics across contrasting lithological settings, such as sedimentary and crystalline aquifers, under different seasonal conditions within the same basin. Furthermore, previous studies have rarely integrated a comprehensive suite of irrigation suitability indices, including SAR, RSC, SSP, MH, KI, MAR, and PI, with multivariate statistical approaches capable of distinguishing lithological controls from evapoconcentration processes. Therefore, the novelty of the present study lies in its integrated and seasonally resolved framework, which simultaneously evaluates surface and groundwater resources while explicitly considering geological heterogeneity, climatic forcing, and microbiological risks. This approach not only addresses an important scientific gap but also provides a transferable methodological framework to support water security and sustainable agricultural management in semi-arid regions worldwide.

As irrigated agriculture continues to expand as an alternative to overcome the climatic constraints of semi-arid regions, the use of low-quality water sources without adequate technical control may intensify irreversible degradation processes, jeopardizing agricultural productivity, food security, and the socio-economic resilience of rural communities. In addition, the alternation between dry and rainy seasons can exert a direct and significant influence on the quality of both surface and groundwater in semi-arid environments. Therefore, the objective of this study is to analyze the physical, chemical, and microbiological parameters of groundwater and surface water sources, as well as to assess irrigation-related indices, during both dry and rainy seasons in the municipality of Iguatu, Ceará, located in the Brazilian semi-arid region.

2 Materials and Methods

2.1 Study Area Description

This study was conducted using surface and groundwater sources in the municipality of Iguatu, located

in the south-central region of Ceará State, Brazil, at 6° 21' 34" S and 39° 17' 55" W, with an altitude of 218 m. According to the Köppen climate classification, the region is hot semi-arid (BSh), with mean annual temperatures of approximately 28 °C and an average total annual precipitation of 996.6 mm. Most rainfall occurs between December and May, while the highest evapotranspiration rates, reaching $1,902.5 \pm 117.8 \text{ mm yr}^{-1}$ are concentrated from August to January (Dubreuil et al., 2019; INMET, 2024; Oliveira et al., 2025).

The Caatinga vegetation is typically described as a xerophytic savanna steppe, dominated by small, thorny trees with twisted trunks, highly branched shrubs, and numerous spiny annual herbs. This biome is predominantly deciduous and displays remarkable adaptations to prolonged drought periods. Its physiognomy is characterized by a dense shrubland with thorny and branched vegetation, frequently interspersed with bromeliads, euphorbias, and cacti (Costa et al., 2024; Nóbrega et al., 2024). Representative species include *Piptadenia stipulacea* (jurema branca), *Mimosa tenuiflora* (jurema preta), *Desmanthus virgatus* (jureminha), and *Mimosa caesalpiniiifolia* (sabiá), which together form a deciduous thorny scrub typical of this semiarid landscape (Nóbrega et al., 2024).

The geomorphology of the area consists of erosional surfaces bordered by crystalline massifs and the margins of sedimentary basins (Silva & Maia, 2017). The crystalline basement features rocky outcrops, while the sedimentary basin includes floodplains. Most agricultural and irrigated semi-intensive and extensive livestock areas are in the sedimentary basin (Oliveira et al., 2025). Geologically, part of the municipality lies within the Iguatu sedimentary basin, covering a total area of 715 km², which also encompasses smaller basins such as Malhada Vermelha and Lima Campos. Lithology is dominated by sedimentary rocks, largely associated with the Mesozoic basin and overlain by more recent sediments. These consist mainly of rhythmites, siltstones, shales, and fine sandstones—predominantly red and green, with interbedded limestones and marls, especially in the Moura Formation and alluvial deposits (FUNCEME, 2024; Lima et al., 2019; Pinéo et al., 2020). Another portion of the municipality lies within the Precambrian crystalline basement, most prominent in the upper Jaguaribe River basin, where metamorphic and igneous rocks, particularly granites and gneisses, occur within the

Ceará Central, Jaguaribeano, and Granjeiro geological domains (FUNCEME, 2024; Silva & Maia, 2017).

Hydrologically, Iguatu is part of the Upper Jaguaribe River sub-basin, whose drainage is dominated by the Jaguaribe River and its main tributaries, including the Trussu River and local streams such as Carnaúba and Antônio (Oliveira et al., 2013; Oliveira et al., 2025). The Orós Reservoir, an artificial water body impounded by the Orós Dam, one of the largest strategic water storage infrastructures in northeastern Brazil, was referred to throughout the manuscript as the “Orós Reservoir”. The Orós Reservoir, located downstream of Iguatu and with a storage capacity of approximately $1.94 \times 10^9 \text{ m}^3$, regulates surface water flow and plays a crucial role in water supply and irrigation across the region. The smaller Trussu Reservoir, with a capacity of approximately $3.01 \times 10^8 \text{ m}^3$, supplements the local water supply, storing water for urban and agricultural use in Iguatu (Dantas et al., 2011). These reservoirs contribute to the regulation of hydrological regimes, mitigate extreme events such as floods and droughts, and support water security, particularly under conditions of high rainfall variability. Groundwater supply is complemented by wells, highlighting the importance of subsurface water to Iguatu. Notably, some wells yield over $80 \text{ m}^3 \text{ h}^{-1}$ of high-quality water, underscoring the resilience of alluvial and fissured aquifer systems as strategic water sources during dry periods, and serving as a key pillar for integrated water resource management in the municipality (Cavalcante et al., 2007).

2.2 Selection of Sampling Sites and Water Collection

The selection of water sampling sites was based on their importance to the irrigation of annual crops, fruit orchards, and pastures, thereby ensuring representative coverage of the municipality. Several sampling points also serve multiple purposes, including livestock watering, recreational activities, and domestic use. To adequately capture the hydrogeochemical heterogeneity of the region, a multi-criteria sampling strategy was adopted, considering: (i) spatial coverage, encompassing upstream, urban, and downstream reaches of the Jaguaribe River, as well as the Orós Reservoir, to represent major surface water bodies; (ii) hydrogeological representativeness, including wells located within both the sedimentary basin (alluvial and fissured aquifers) and the Precambrian

crystalline basement (fractured aquifers), enabling comparisons among distinct lithological settings and their influence on water quality; (iii) agricultural relevance, prioritizing areas intensively used for the irrigation of annual crops (Poaceae and Fabaceae), fruit orchards (banana and guava), and pastures, which constitute the predominant agricultural activities in the municipality; and (iv) logistical accessibility and landowner authorization, ensuring the feasibility of sampling during both seasonal campaigns. This integrated approach was designed to characterize the spatial variability of water resources used in local agriculture, thereby enhancing the study's reproducibility and extending the applicability of its findings to comparable semi-arid environments.

Surface and groundwater samples were collected during the rainy season (May 2023) and the dry season (August 2023). Thirteen sampling sites were established, comprising eight groundwater locations associated with wells and five surface water sites, including

one point in the Orós Reservoir and four sites distributed along the Jaguaribe River (Fig. 1). Additionally, hydrogeological, geomorphological, and elevation maps were generated to improve the spatial characterization of the study area and to provide a better understanding of the distribution and environmental setting of the sampling sites (Fig. 2). At each site, six independent field replicates were collected within a short sampling interval (2–3 h) and distributed within a radius of up to 20 m from the central point to account for small-scale spatial variability. Each replicate was analyzed individually for physicochemical and microbiological parameters, and the results are presented as mean values $\pm$ standard error of the mean (SEM) in the supplementary material. These replicates represent true field replicates rather than subsamples from a composite sample or analytical repetitions, providing a more realistic representation of in situ heterogeneity and increasing the statistical robustness of seasonal comparisons. Groundwater sampling depths ranged

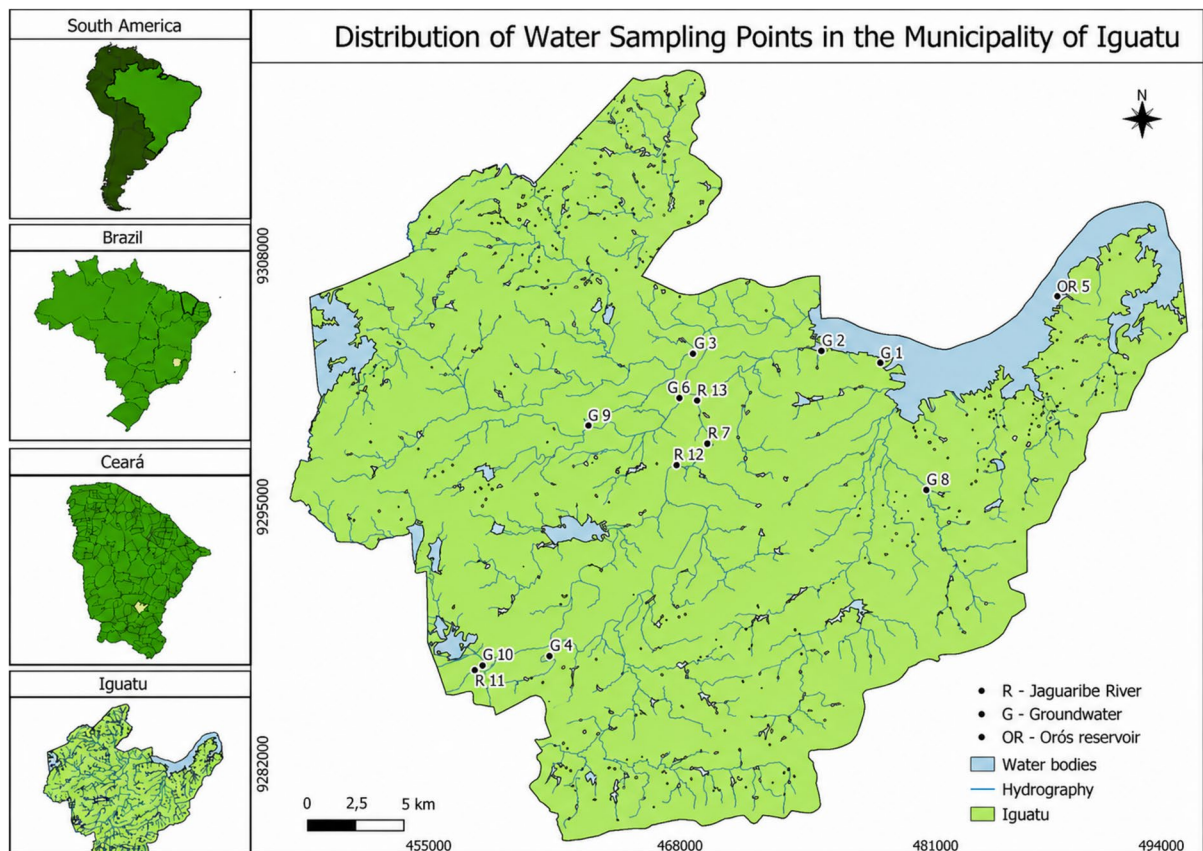


Fig. 1 Location map of groundwater and surface water sampling points in the municipality of Iguatu, Ceará, Brazil

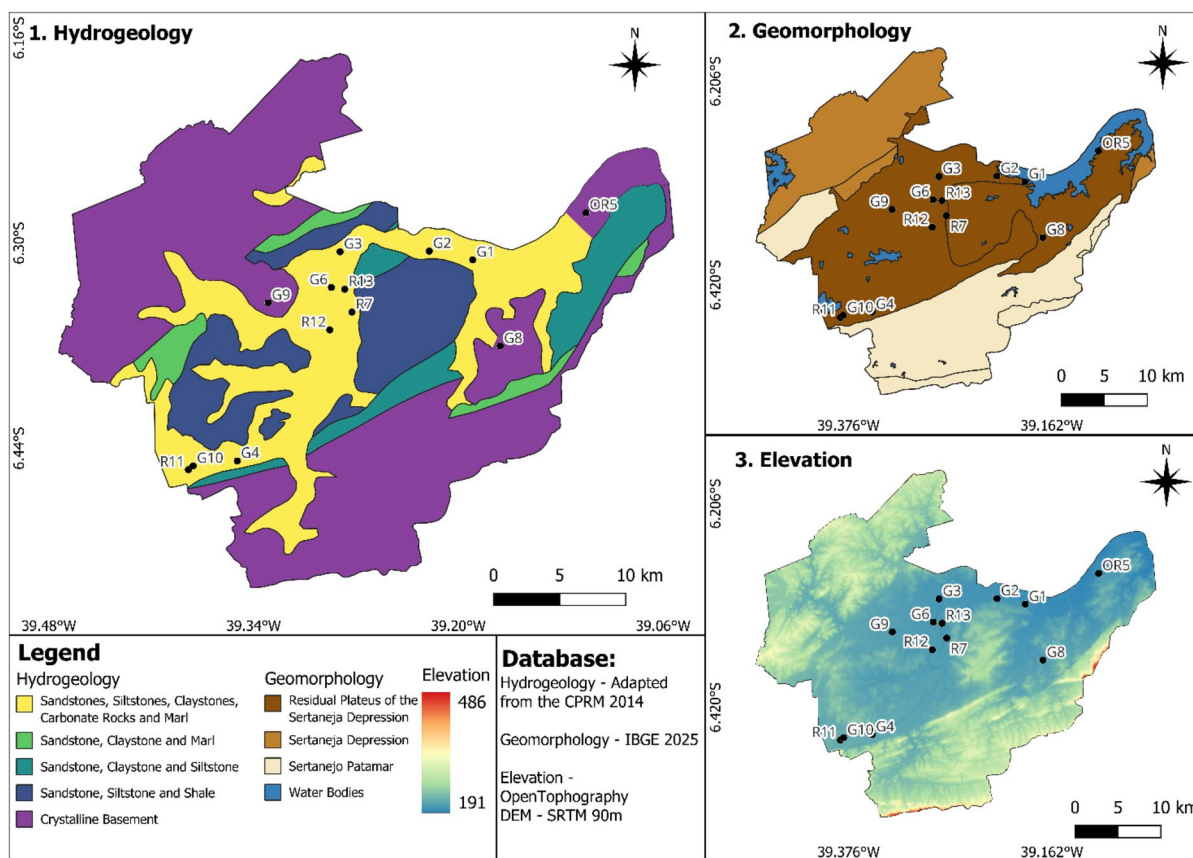


Fig. 2 Hydrogeological, geomorphological, and elevation maps of groundwater and surface water sampling points in the municipality of Iguatu, Ceará, Brazil

from 25 to 70 m, corresponding to shallow wells. Due to methodological constraints, it was not possible to determine the exact screened intervals or filter depths for individual wells. Consequently, in-depth information was derived from field observations and reports from local farmers and water users.

Historical land use and geological characteristics of the sampling sites are presented in Table 1. Detailed information on recent land use and land cover dynamics, including agricultural expansion, livestock intensification, and forest degradation within the study area, can be found in Oliveira et al. (2025), which provides a comprehensive assessment of current landscape conditions and anthropogenic pressures in the Upper Jaguaribe Basin.

Water samples for physicochemical analyses were collected in 500 mL polyethylene bottles, which were triple rinsed with water from the respective sampling source to ensure container homogenization.

Groundwater samples from wells were collected after opening the outlet and allowing the water to flow for three minutes. Surface water samples from the river or reservoir were collected 2–3 m from the banks of the water bodies. For microbiological analyses, surface and groundwater samples were collected in sterilized 70 mL plastic containers, stored in insulated coolers with ice, and kept under refrigeration until analysis.

2.3 Physical, Chemical, and Microbiological Analyses of Water, and Calculated Indices

The parameters analyzed included pH, electrical conductivity (EC), and turbidity, measured using a benchtop pH meter, conductivity meter, and turbidimeter, respectively. Potassium and sodium concentrations were determined by flame photometry. Calcium, magnesium, chloride, carbonate, and bicarbonate were analyzed by titrimetry, and sulfate by turbidimetry. All physicochemical

Table 1 Sampling sites, geological structure, and historical use of surface and groundwater in the municipality of Iguatu, Ceará, Brazil

Site	Water type	Geological structure	Historical use
G1*	Groundwater	Sedimentary basin	Agricultural use (pasture irrigation) and livestock watering
G2	Groundwater	Sedimentary basin	Agricultural use (irrigation of annual crops—Poaceae and Fabaceae). During drought, this well serves as an alternative source for domestic human consumption
G3	Groundwater	Sedimentary basin	Agricultural use (irrigation of annual crops—Poaceae, Fabaceae, and Cucurbitaceae)
G4	Groundwater	Sedimentary basin	Agricultural use (irrigation of fruit crop—banana)
OR5	Surface water	Crystalline basement	Five collection points in the Orós Reservoir were composited into a single sample for analysis with six replicates. These waters are used for irrigating various crops and livestock watering. Also used for human water supply
G6	Groundwater	Sedimentary basin	Agricultural use (irrigation of pastures and annual crops—Poaceae and Fabaceae)
R7	Surface water	Sedimentary basin	Jaguaribe River (irrigation of various crops, including agroforestry systems, near a Permanent Preservation Area—PPA)
G8	Groundwater	Crystalline basement	Agricultural use (pasture irrigation)
G9	Groundwater	Crystalline basement	Agricultural use (irrigation of sugarcane, fruit crops – guava, and vegetables)
G10	Groundwater	Sedimentary basin	Agricultural use (irrigation of fruit crop—banana)
R11	Surface water	Sedimentary basin	Jaguaribe River (irrigation of various crops and recreational human use). Upstream from the municipality of Iguatu
R12	Surface water	Sedimentary basin	Jaguaribe River (irrigation of various crops and recreational human use). Within the urban area of Iguatu
R13	Surface water	Sedimentary basin	Jaguaribe River (irrigation of various crops and recreational human use). Downstream from the municipality of Iguatu

*These numbers indicate the order in which the water samples were collected

analyses were conducted at the Laboratory of Soil, Water, and Plant Analysis (LASAP/UFERSA) following the Standard Methods for the Examination of Water and Wastewater (Baird et al., 2017).

For metal determination, water samples underwent pre-treatment by adding 3 mL of nitric acid (HNO₃) to each 100 mL of sample. The mixture was heated on a hot plate until the volume was reduced to approximately 50 mL, then brought back to 100 mL in a volumetric flask, following APHA (2005) recommendations. Concentrations of copper (Cu), manganese (Mn), iron (Fe), zinc (Zn), chromium (Cr), cadmium (Cd), and lead (Pb) were quantified using atomic absorption spectrophotometry.

Microbiological analyses were performed at the Laboratory of Inspection of Products of Animal Origin (LIPOA/UFERSA) according to the methodology established by Normative Instruction No. 30/2018 of the Ministry of Agriculture, Livestock, and Food Supply (MAPA) (Brazil, 2018). The presence of total coliforms, thermotolerant coliforms, and *Escherichia coli* was determined following ISO 9308–1, and thermotolerant coliforms were analyzed according to ISO 6222.

For the calculation of irrigation water quality indices, the following chemical elements were considered: calcium (Ca²⁺), magnesium (Mg²⁺), sodium (Na⁺), potassium (K⁺), chloride (Cl[−]), bicarbonate (HCO₃[−]), and carbonate (CO₃^{2−}), all expressed in mmol/L^{−1}. The calculated indices included: Sodium Absorption Ratio (SAR), Magnesium Adsorption Ratio (MAR), Magnesium Hazard (MH), Residual Sodium Carbonate (RSC), Soluble Sodium Percentage (SSP), Kelly's Index (KI), and Permeability Index (PI), as described in Eqs. 1 through 7, respectively.

$$SAR = \frac{Na^+}{\sqrt{\frac{Ca^{2+} + Mg^{2+}}{2}}} \quad (1)$$

$$MAR = \frac{Mg^{2+}}{\sqrt{\frac{Ca^{2+} + Mg^{2+} + Na^+}{2}}} \quad (2)$$

$$MH = \frac{Mg^{2+}}{Ca^{2+} + Mg^{2+}} \times 100 \quad (3)$$

$$RSC = (CO_3^{2-} + HCO_3^-) - (Ca^{2+} + Mg^{2+}) \quad (4)$$

$$SSP = \frac{Na^+ + K^+}{Ca^{2+} + Mg^{2+} + Na^+ + K^+} \times 100 \quad (5)$$

$$KI = \frac{Na^+}{Ca^{2+} + Mg^{2+}} \times 100 \quad (6)$$

$$PI = \frac{Na^+ + \sqrt{HCO_3^-}}{Ca^{2+} + Mg^{2+} + Na^+} \times 100 \quad (7)$$

The analytical quality of the hydrochemical dataset was assessed using the Charge Balance Error (CBE). The percentage error was computed according to the methodology proposed by Freeze and Cherry (1979) and Kehew (2001), as expressed in Eq. (8):

$$\text{Charge Balance Error (CBE)}(\%) = \left(\frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \right) \times 100 \quad (8)$$

where the ion concentration unit is in milliequivalent per liter (meq L⁻¹).

All samples exhibited Charge Balance Error (CBE) values ranging from 5 to 10%, which are considered acceptable for hydrochemical studies (Rahman et al., 2021). Supplementary Table S1 was included to provide a concise assessment of analytical quality control.

Hydrogeochemical controls on water composition were further explored using Gibbs diagrams (Gibbs, 1970). The ratios $Na^+/(Na^+ + Ca^{2+})$ and $Cl^-/(Cl^- + HCO_3^-)$ were plotted as a function of total dissolved solids (TDS) on a logarithmic scale to generate the cationic and anionic Gibbs diagrams, respectively, allowing discrimination among the relative influences of atmospheric precipitation, rock weathering, and evaporation-crystallization processes on the evolution of water chemistry (Nguyen et al., 2015).

2.4 Statistical Analysis

Multivariate statistical methods were employed to distinguish between seasons and water sources. To build a robust analytical matrix, Pearson's correlation and linear regression analyses ($p \leq 0.05$) were first applied to identify interdependent variables of relevance to the dataset. Subsequently, Principal Component Analysis (PCA) and cluster analyses were performed using the R statistical environment. Salinity and sodicity risks were evaluated using the U.S. Salinity Laboratory Diagram (Richards et al., 1954), while ionic water classification was conducted with the Piper Diagram. Both diagrams were generated

using Qualigraf software, version 1.17 (FUNCEME, 2024).

3 Results

3.1 Pearson's Correlation, Principal Component Analysis, Nutrient and Metal Contents

Pearson's linear correlation matrix for the rainy and dry seasons revealed a significant dynamic in the interactions among water chemical variables, reflecting the impact of seasonality. In the dry season, statistically significant correlations were observed between major cations and anions: Mg^{2+} and HCO_3^- ($r=0.81^*$), Ca^{2+} and HCO_3^- ($r=0.80^{***}$), Fe^{2+} and SO_4^{2-} ($r=0.79^{**}$), as well as Na^+ and Cl^- ($r=0.85^*$) and Na^+ and HCO_3^- ($r=0.66^*$), suggesting a strong influence of carbonate mineral dissolution and soluble salts (Fig. 3A). Total hardness showed a robust positive correlation with Mg^{2+} ($r=0.86^*$), Ca^{2+} ($r=0.90^{***}$), and HCO_3^- ($r=0.90^{***}$), while SAR correlated with Cl^- ($r=0.90^*$) in addition to sodium, reinforcing the role of Na^+ as a key driver of sodicity risk. Turbidity correlated significantly with Fe^{2+} ($r=0.98^{***}$) and SO_4^{2-} ($r=0.79^{**}$), indicating the influence of suspended particles associated with reduced hydrological activity during the dry season (Fig. 3A).

In the rainy season, similar correlation patterns were observed, suggesting that comparable hydrochemical controls remain active under different seasonal conditions. Although rainfall-driven

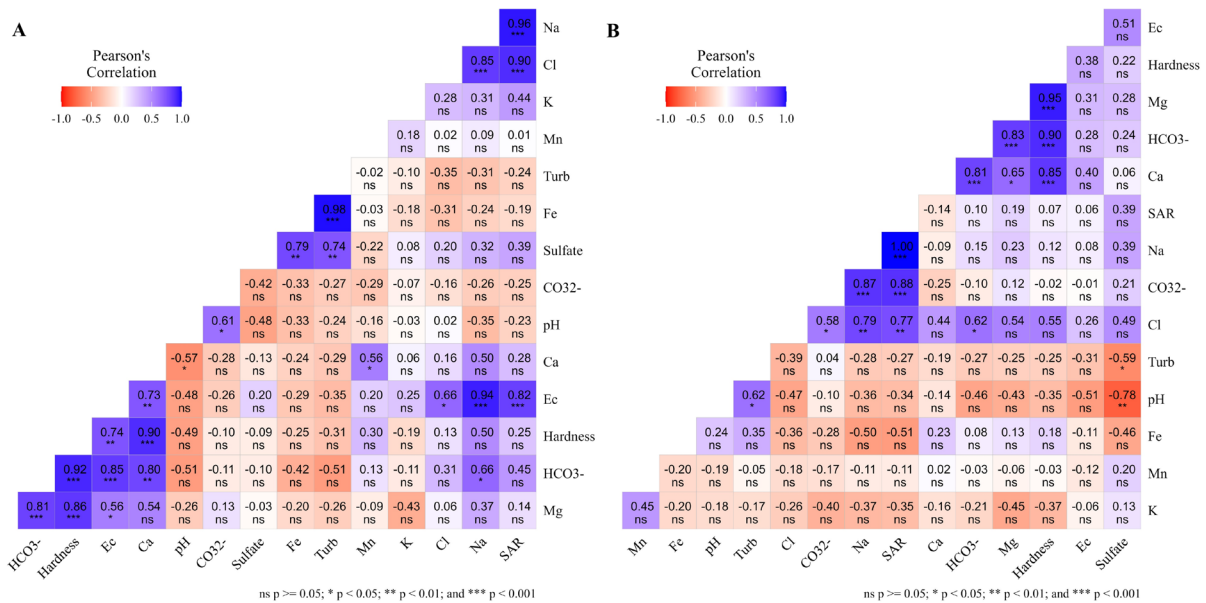


Fig. 3 Pearson's linear correlation matrix between the physical and chemical parameters of water from groundwater wells (G1-G4, G6, G8, G9, G10), surface reservoir (OR5—Orós Reservoir), and stretches of the Jaguaribe River (R7, R11, R12,

R13). **A)** Dry season; **B)** Rainy season. Significant correlations $r \geq 0.70$ (***, **, and * denote significance at 0.1%, 1%, and 5%, respectively)

recharge likely promotes dilution processes, as evidenced by the reduction in salinity-related parameters at several sampling sites, the persistence of significant associations among major ions indicates that water–rock interactions continue to exert an important influence on groundwater and surface water chemistry throughout the hydrological cycle. Associations between Mg^{2+} and HCO_3^- ($r = 0.83^*$), Mg^{2+} and Ca^{2+} ($r = 0.65^*$), and Ca^{2+} and HCO_3^- ($r = 0.81^{***}$) persisted, indicating that carbonate dissolution processes remain active during aquifer recharge. Correlations between Na^+ and Cl^- ($r = 0.79$) and Na^+ and HCO_3^- ($r = 0.87^{***}$) reinforced the same trend observed in the dry season, suggesting a dominant ionic control of these chemical species. Hardness remained strongly associated with Mg^{2+} , Ca^{2+} , and HCO_3^- , with even higher coefficients ($r \geq 0.85^{***}$). SAR also correlated significantly with Cl^- ($r = 0.77^*$) and HCO_3^- ($r = 0.88^{***}$), in addition to sodium, maintaining the sodicity risk pattern (Fig. 3B).

For the Principal Component Analysis (PCA), data were normalized in advance and analyzed based on the first two principal components (PC1 and PC2). PCA clearly distinguished sample groups

according to water source type and seasonality. In the dry season, PC1 (39.6%) and PC2 (21.5%) together explained 61.1% of the total variance. Wells G9 and G6 showed a strong negative association with PC1, indicating salinization (Fig. 4A). In contrast, samples R11, R12, and R13 consistently clustered in the positive quadrant of PC1, reinforcing that surface waters maintain lower salt concentrations and fewer adverse chemical parameters even during drought. EC, HCO_3^- , hardness, and Mg^{2+} remained determinant in PC2, contributing to the differentiation of harder groundwater, such as G8.

In the rainy season, PC1 (36.6%) and PC2 (24.5%) jointly explained 61.1% of the total variance, like the dry season. PC1 was strongly positively correlated with pH, K^+ , Mn, and Fe, and negatively correlated with Na^+ , CO_3^{2-} , SAR, Cl^- , SO_4^{2-} , and TDS. PC2 was more closely associated with hardness, Ca^{2+} , Mg^{2+} , and HCO_3^- , indicating local geochemical (likely lithological) influence. Samples R7, R11–R13, OR5, and G1, G2, and G4 clustered in the positive quadrant of PC1, indicating better water quality conditions associated with higher pH, K^+ , and Mn-typical of more oxygenated, less saline surface waters. Conversely, G9 samples were strongly associated

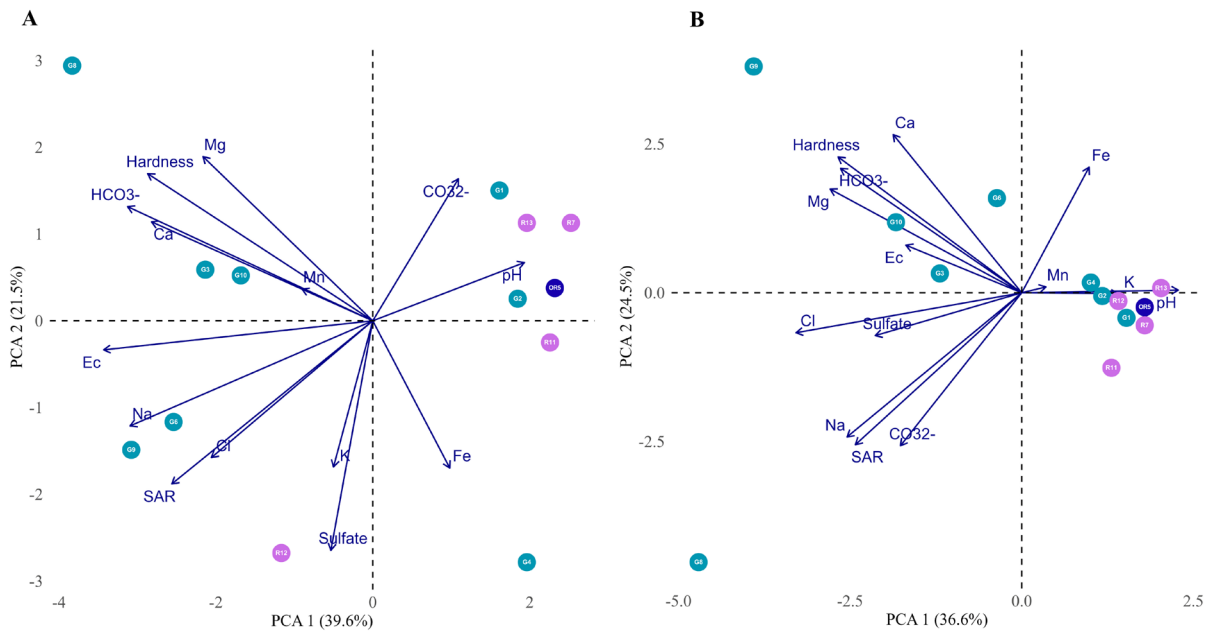


Fig. 4 Principal component analysis (PCA) and distribution of zones in the multivariate space of physical and chemical parameters of water from groundwater wells (G1–G4, G6, G8,

G9, G10—green), surface reservoir (OR5—Orós Reservoir—blue), and stretches of the Jaguaribe River (R7, R11, R12, R13—purple). **A)** Dry season; **B)** Rainy season

with the negative quadrant of PC1, suggesting salinity, and G8 indicated high salt content linked to hardness (Na^+ , CO_3^{2-} , SAR, Cl^-), differing from the dry season (Fig. 4B).

The positive correlations among Ca^{2+} , Mg^{2+} , and HCO_3^- observed in both seasons indicate that water chemistry is strongly controlled by water–rock interactions, particularly the dissolution of carbonate-bearing materials associated with the Moura Formation and recent alluvial deposits. The infiltration of CO_2 -enriched recharge water through these lithological units promotes the dissolution of calcite and dolomite, releasing Ca^{2+} , Mg^{2+} , and HCO_3^- into solution and sustaining the hardness–alkalinity relationships identified in the correlation matrix. In contrast, the marked hydrochemical differentiation observed among groundwater samples, especially the higher salinity, hardness, and Na^+ and Cl^- concentrations recorded in wells G8 and G9, appear to be associated with their occurrence within the Precambrian crystalline basement. In these fractured granitic and gneissic aquifers, prolonged water–rock interaction and the weathering of feldspars, micas, and ferromagnesian minerals contribute to the enrichment of dissolved ions, while low primary porosity and limited

hydraulic conductivity reduce flushing processes and favor solute accumulation. Conversely, wells located within the sedimentary basin (e.g., G1, G2, and G4) exhibited lower salinity levels and softer waters, reflecting higher permeability, enhanced recharge conditions, and hydrochemical compositions influenced by alluvial inputs and carbonate equilibrium. This lithological contrast is further supported by the PCA results, in which PC1, strongly associated with Na^+ , Cl^- , SAR, and EC, discriminates groundwater influenced by crystalline basement environments, whereas PC2, dominated by Ca^{2+} , Mg^{2+} , HCO_3^- , and hardness, reflects the pervasive influence of carbonate dissolution processes throughout the study area. Together, these findings highlight the fundamental role of geological setting in shaping the spatial variability and hydrochemical evolution of water resources in the Upper Jaguaribe Basin.

Chemical element concentrations for the different water sources in the dry and rainy seasons are presented in Supplementary Table S2. Overall, in the dry season, mean EC, Na^+ , Ca^{2+} , Mg^{2+} , Cl^- , SO_4^{2-} , and alkalinity (HCO_3^-) values were higher in several samples (notably G6, G8, and G9), reflecting increased salinity due to evapoconcentration typical of the

season (Table S2). In the rainy season, although some sites, especially G8, G9, and G10, still maintained elevated concentrations, precipitation-driven dilution reduced salinity levels in most samples, particularly for Na^+ , Cl^- , and SO_4^{2-} . pH values ranged from 6.6 (G4) to 9.6 (R7) in the dry season and from 6.6 (G10) to 8.1 (R13) in the rainy season, without exceeding recommended limits. Turbidity showed greater variation in the dry season at sites such as G4 and R12, suggesting increased suspended solids linked to anthropogenic activity or erosive processes. TDS concentrations followed the EC pattern, with maximum values of $\sim 0.34 \text{ g L}^{-1}$ (R12) in the dry season and 0.70 g L^{-1} (G10) in the rainy season. Supplementary Tables S3 and S4 present nutrient comparison parameters according to World Health Organization guidelines and irrigation reference values (Ayers & Westcot, 1985), highlighting sites that exceeded recommended thresholds.

Supplementary Table S5 presents micronutrient and metal concentrations (Cu, Mn, Fe, Zn, Cr, Ni, Cd, Pb). During the dry season, Fe levels exceeded the World Health Organization ((2019) limit of 0.3 mg L^{-1} in several locations, most notably in well G4, which showed an extreme value of $5.738 \pm 0.221 \text{ mg L}^{-1}$, suggesting localized contamination. Wells G2 (0.788 mg L^{-1}) and G10 (0.405 mg L^{-1}) also exceeded this threshold, potentially posing risks of irrigation system clogging due to ferric precipitation. For Mn, with a World Health Organization (2019) limit of 0.4 mg L^{-1} , exceptionally high concentrations were observed in G3 ($1.650 \pm 0.043 \text{ mg L}^{-1}$) and, to a lesser extent, in G4 (0.052 mg L^{-1}) and G10 (0.071 mg L^{-1}). Although Cu was detected at all sites, concentrations remained below the limit of 2.0 mg L^{-1} ; the same applied to Zn, which was detected only in trace amounts and within the safe range ($< 3 \text{ mg L}^{-1}$). Cd, Pb, Ni, and Cr were not detected in most samples, except for traces of Ni and Cr in R12, both below critical thresholds (Cr: 0.05 mg L^{-1} ; Ni: 0.07 mg L^{-1}).

In the rainy season, a general dilution trend for metals was observed, with significant reductions in Fe and Mn concentrations at most sites. However, wells G3 and G4 continued to present elevated Mn levels (1.167 and 1.210 mg L^{-1} , respectively), again exceeding WHO limits and possibly related to metal mobilization due to changes in the water table (Supplementary Table S4). Although Fe levels decreased,

they remained high in G1 (0.200 mg L^{-1}) and R7 (0.213 mg L^{-1}), close to the WHO threshold. Cu remained stable and below critical values (Supplementary Table S4). Ni and Cr reappeared in low concentrations (e.g., OR5 and G4), but always within international standards. No Pb or Cd was detected in any sample, indicating a low probability of direct anthropogenic contamination by these elements during the study period.

3.2 Irrigation Water Quality Indices

An integrated analysis of irrigation water quality indices revealed that water quality varied among the different sources and between sampling periods; however, some indices were more sensitive than others. For the Sodium Adsorption Ratio (SAR) and Residual Sodium Carbonate (RSC), all sites in both periods remained well below the critical thresholds for irrigation ($\text{SAR} < 10$ and $\text{RSC} < 2.5$), indicating low risk of sodification and carbonation (Table 2). The only exception was R11 during the rainy season, with $\text{RSC} = 1.23$, which is at the marginal threshold. For the Soluble Sodium Percentage (SSP), values were generally higher during the dry season compared to the rainy season, particularly for R12 (54.3%) and G9 (48.1%), both near the threshold ($\text{SSP} < 50$ —acceptable). In the rainy season, all sites recorded $\text{SSP} < 50\%$, suggesting improved irrigation suitability (Table 2).

For Magnesium Hazard (MH), most groundwater and surface water sites exceeded 50% during the dry season, except for G3, indicating potential risks of compaction and reduced infiltration in clayey soils. In the rainy season, most MH values were $< 50\%$ (acceptable), except for G8, G9, and OR5. During the dry season, all sites exhibited Magnesium Adsorption Ratio ($\text{MAR} > 1$), indicating Mg^{2+} dominance over Ca^{2+} , which can exacerbate soil dispersion under continuous irrigation; during the rainy season, dilution effects were observed only at G1, G4, R11, and R13. For Kelly's Index (KI), in the dry season, only G9 (0.90) and R12 (0.97) approached the threshold ($\text{KI} < 1$, good quality). In the rainy season, all points showed very low KI values (< 0.76), indicating good water quality. Finally, for the Permeability Index (PI), all sites in both seasons remained within the moderate ($\text{PI} = 25\text{--}75\%$) to excellent ($\text{PI} > 75\%$) range for irrigation water quality (Table 2).

Table 2 Water quality indices for irrigation of groundwater (G) and surface water (R-river and OR- Orós Reservoir) sources in the municipality of Iguatu-CE, Brazil

Sites	Dry period						
	SAR	RSC	SSP	MH	MAR	KI	PI
G1	1.01	0.29	30.6%	56.2%	1.27	0.38	65.8%
G2	1.00	−0.27	28.5%	60.3%	1.48	0.35	60.1%
G3	1.59	−0.42	36.2%	43.4%	1.13	0.50	61.1%
G4	1.23	−0.83	34.0%	63.3%	1.32	0.48	64.0%
OR5	0.57	−0.13	21.7%	65.4%	1.58	0.22	59.4%
G6	2.26	0.37	45.4%	51.6%	1.18	0.75	70.1%
R7	1.58	0.62	41.0%	71.5%	1.40	0.63	67.8%
G8	1.70	0.72	32.2%	60.9%	1.81	0.47	59.5%
G9	2.64	1.11	48.1%	60.3%	1.28	0.90	75.6%
G10	1.49	−0.17	33.7%	50.8%	1.33	0.47	61.3%
R11	1.05	0.22	31.9%	63.3%	1.33	0.42	69.9%
R12	2.55	0.55	54.3%	57.0%	1.06	0.97	77.9%
R13	1.40	0.15	37.5%	60.2%	1.24	0.55	70.6%
Sites	Rainy period						
	SAR	RSC	SSP	MH	MAR	KI	PI
G1	0.13	−0.07	6.2%	31.1%	0.69	0.06	63.6%
G2	0.12	1.01	5.6%	45.8%	1.07	0.05	70.5%
G3	0.19	0.21	6.5%	42.6%	1.23	0.06	51.6%
G4	0.12	0.40	6.5%	46.5%	0.97	0.06	73.0%
OR5	0.11	0.74	5.7%	53.2%	1.12	0.05	76.8%
G6	0.27	0.80	9.5%	45.3%	1.19	0.10	60.3%
R7	0.12	−0.36	5.3%	47.8%	1.15	0.05	56.3%
G8	0.19	−0.47	4.9%	58.1%	2.11	0.05	39.7%
G9	2.00	0.35	43.2%	56.6%	1.13	0.76	75.0%
G10	0.12	−0.18	4.6%	45.2%	1.28	0.04	50.0%
R11	0.15	1.23	6.9%	19.1%	0.42	0.07	77.3%
R12	0.09	−0.29	3.9%	42.9%	1.07	0.03	54.9%
R13	0.10	0.58	5.1%	40.5%	0.91	0.05	69.7%

*SAR Sodium Absorption Ratio, RSC Residual Sodium Carbonate, SSP Soluble Sodium Percentage, MH Magnesium Hazard, MAR Magnesium Adsorption Ratio, KI Kelly's Index, PI Permeability Index, G Groundwater, OR Oros reservoir, R River

3.3 Piper and Gibbs Diagrams, CONAMA, U.S.S.L Classifications, and Hierarchical Cluster Analysis (HCA)

The combined analysis of hydrochemical data using Piper Diagram for the dry and rainy seasons characterized the dominant typologies of water sources in municipality of Iguatu. The distribution of samples in the cation and anion triangles of the Piper diagram revealed marked differences between water source types and the significant influence of climatic seasonality on ionic composition. Groundwater (G) was generally characterized during the dry season by mixed types dominated by Na^+ , K^+ , Ca^{2+} , and Mg^{2+} as major cations, with Cl^- and HCO_3^- as predominant anions, classifying them as mixed chloride or mixed

bicarbonate waters, typical of environments subject to intense evapoconcentration, low recharge rates, and ion-exchange processes (Fig. 5A). In the rainy season, partial dilution of dissolved salts shifted the samples toward the left side of the diagram, reflecting mixed bicarbonate types with varying proportions of Ca^{2+} and Mg^{2+} (Fig. 5B). This seasonal transition indicates a relative reduction in salinity and changes in ionic regimes associated with rainfall recharge and the influence of varied geological materials in the aquifer.

Orós Reservoir exhibited intermediate characteristics between groundwater and river water. In the dry season, its composition was slightly more saline, with moderate proportions of magnesium and bicarbonate, reflecting concentration effects from evaporation and possible upstream ionic contributions (Fig. 5A).

In the rainy season, the composition shifted toward mixed bicarbonate waters, indicating more diluted conditions and direct influence from surface runoff (Fig. 5B). This pattern reinforces the reservoir's role as an accumulation and mixing zone, with quality strongly modulated by seasonality and the origin of inflowing waters.

Water from the Jaguaribe River showed varying classifications depending on the season and sampling location. In the dry season, most points were mixed-type for cations, except for R7, which was sodic, yielding mixed bicarbonate types (R13 and R11), mixed chloride (R7), and sodic mixed (R12) waters (Fig. 5A). In the rainy season, dilution was evident, with predominance of calcium bicarbonate types, except for R7, which was classified as mixed bicarbonate (Fig. 5B).

The Gibbs diagrams indicate that most samples are concentrated within the rock-weathering dominance field, demonstrating that water–rock interaction is the primary mechanism controlling hydrogeochemical evolution in the Upper Jaguaribe Basin (Fig. 6). Groundwater samples associated with sedimentary aquifers were predominantly distributed within this domain, reflecting the influence of carbonate dissolution and silicate weathering processes. In contrast, samples derived from the crystalline basement, particularly during the dry season, shifted toward the evaporation–crystallization dominance field, accompanied by higher TDS values and increased $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ ratios. Surface waters from the Jaguaribe River and the Orós Reservoir occupied intermediate positions between the precipitation and rock-weathering domains, suggesting the combined influence of seasonal recharge processes and lithological controls. Overall, the seasonal displacement of samples toward more saline conditions during the dry period indicates that evapoconcentration acts as a secondary mechanism for solute enrichment, without substantially modifying the dominant hydrogeochemical signatures of the basin.

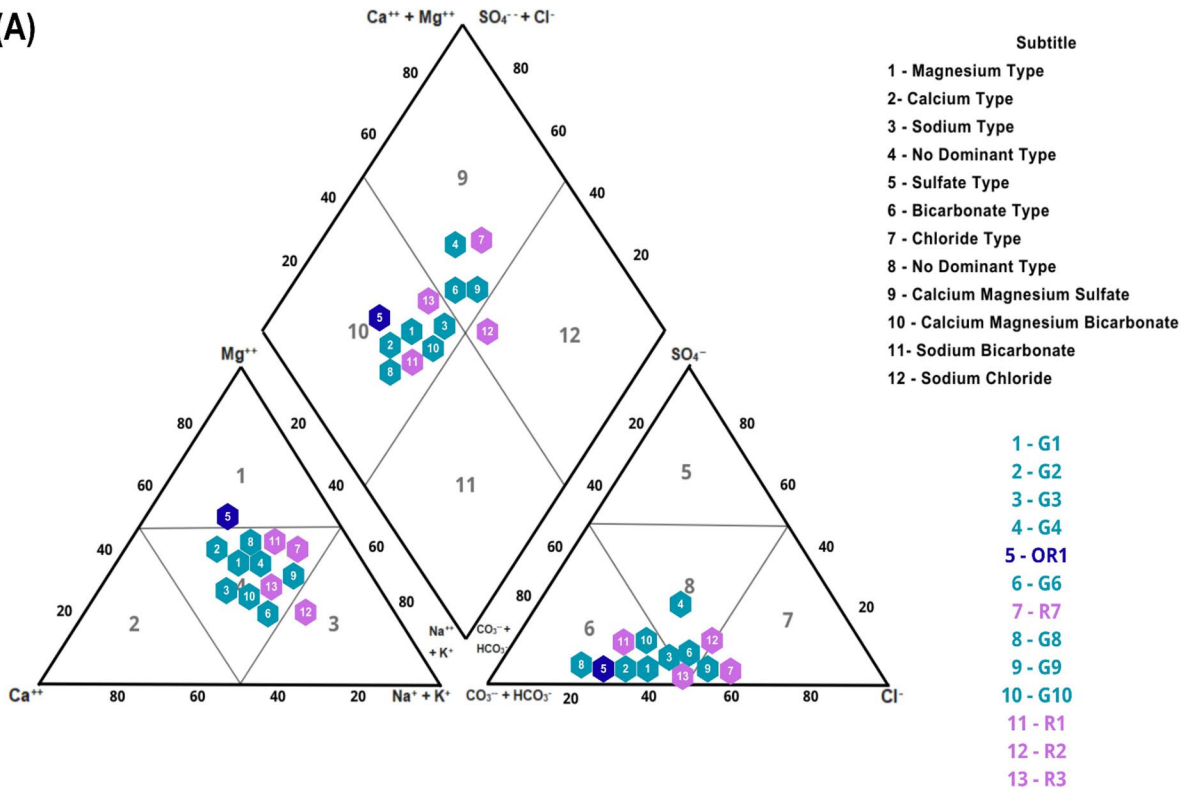
Physicochemical quality assessment based on electrical conductivity (EC) and total dissolved solids (TDS) allowed salinity classification following CONAMA Resolution No. 357/2005 (Table 3). In the dry season, five of the eight wells (G3, G6, G8, G9, and G10) exceeded 500 mg L^{-1} TDS, thus classified as brackish (Table 3). Conversely, G1, G2, and G4

remained freshwater despite high EC. In the rainy season, significant dilution occurred in six wells, shifting most to freshwater classification, including G3 and G6, previously brackish. Only G8 and G9 remained brackish, with $\text{TDS} > 600 \text{ mg L}^{-1}$, suggesting persistent salinity from geological sources. The Orós Reservoir maintained freshwater status in both seasons, with TDS of 201.5 mg L^{-1} (dry) and 169.0 mg L^{-1} (rainy). Jaguaribe River waters were generally freshwater in both seasons, except R13 in the dry period ($\text{TDS} = 546 \text{ mg L}^{-1}$, brackish). In the rainy season, all river sites became freshwater, reflecting dilution by rainfall and efficient hydrological renewal.

The U.S. Salinity Laboratory (USSL) diagram classified sites according to salinity (C1–C4) and sodicity (S1–S4) hazards. In the dry season, six of eight wells fell into C2–S1 or C3–S1 classes, with G6, G8, and G9 showing $\text{EC} > 870 \mu\text{S cm}^{-1}$ and SAR between 1.5–2.66 (C3–S1: high salinity, low sodicity hazard) (Fig. 7A). In the rainy season, EC and SAR decreased across all wells, shifting to lower salinity classes (C2–S1) and, for G4, to C1–S1, indicating recharge response. G8 and G9 remained C3–S1 due to persistent mineralization (Fig. 7B). The Orós Reservoir remained stable (C2–S1) in both seasons ($\text{EC} = 260\text{--}310 \mu\text{S cm}^{-1}$, $\text{SAR} < 0.6$). Most river sites were C2–S1 in the dry season, except R13 (C3–S1, $\text{EC} = 840 \mu\text{S cm}^{-1}$). In the rainy season, all river sites shifted to C2–S1 with reduced SAR and salinity.

Hierarchical Cluster Analysis (HCA) using Euclidean distance and Ward's linkage identified four distinct clusters in both seasons, reflecting seasonal and hydrogeochemical variability among water sources (Fig. 8). In the dry season, Cluster 1 isolated G8 due to high Na^+ , Cl^- , SAR, and EC (indicative of salinization). Cluster 2 grouped G3, R12, G10, G6, and G9, characterized by high hardness, Mg^{2+} , HCO_3^- , and EC (except R12). Cluster 3 isolated G4 due to intermediate hardness and Mg^{2+} . Cluster 4 grouped river and reservoir waters with G1 and G2, reflecting dilution and stable pH, K^+ , and Mn levels (Fig. 8A). In the rainy season, Cluster 1 isolated G9 for persistent high hardness and alkaline salts. Cluster 2 grouped OR5, R7, R12, R13, G2, G1, and R11, indicating seasonal homogenization from aquifer recharge and mixing. Cluster 3 again isolated G8, albeit with less divergence, suggesting lithological control of salinity. Cluster 4 grouped G4, G3, G6, and G10, showing

(A)



(B)

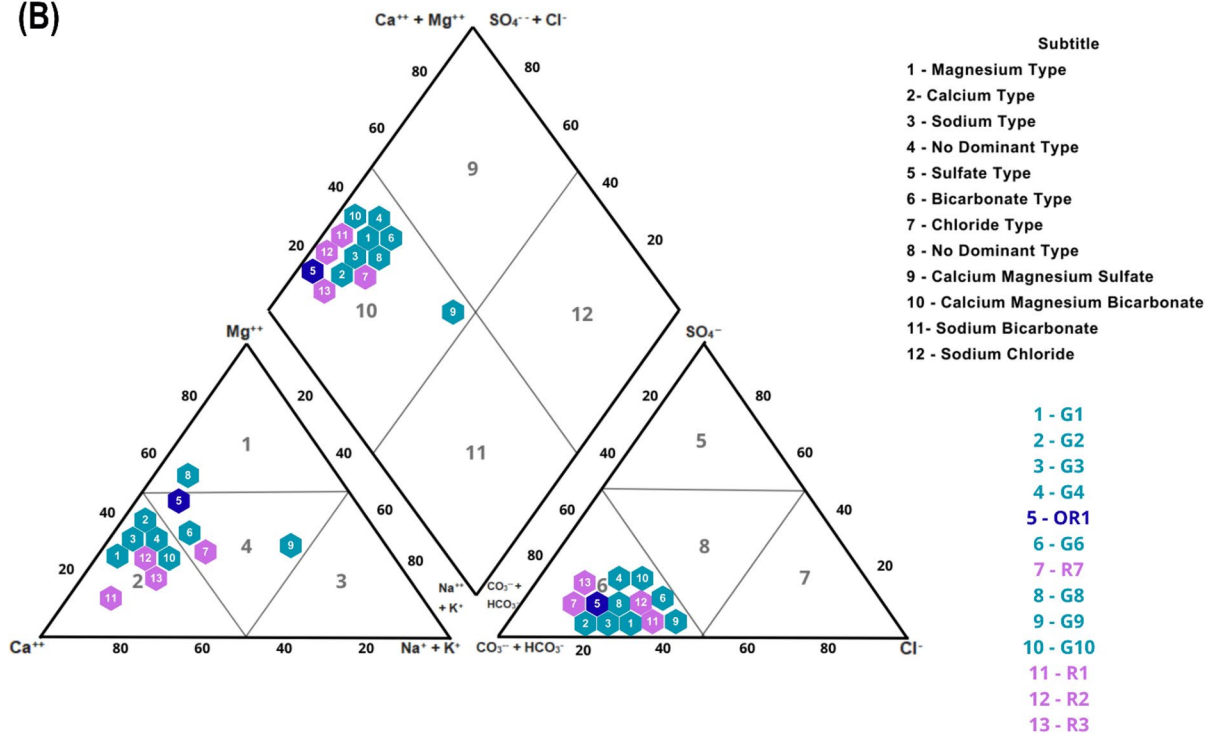


Fig. 5 Water samples collected at points distributed among groundwater wells (G1–G4, G6, G8, G9, G10—green), surface reservoir (OR5—Orós Reservoir—blue), and stretches of the Jaguaribe River (R7, R11, R12, R13—purple), during the dry (A) and rainy (B) periods, in the municipality of Iguatu, Ceará, Brazil, plotted on Piper's trilinear diagram

similar geochemical composition with moderate hardness, Ca^{2+} , HCO_3^- , and EC (Fig. 8B).

The microbiological assessment revealed the presence of total coliforms at varying population level across all sampling points in both the dry and rainy seasons. Notably, sites G3, G4, G8, R7, and R12 recorded the highest counts, ranging from 24 to 46 MPN mL^{-1} . Regarding thermotolerant coliforms—an indicator more specifically associated with recent fecal contamination—sites G3 and R7 also stood out, with values reaching up to 36 MPN mL^{-1} during the dry season, indicating a high risk of enteric pathogen presence (Table 4).

For *Escherichia coli*, a direct indicator of fecal contamination and microbiological risk, most samples showed absence (not detected or <3.0 CFU mL^{-1}). However, concerning levels were detected in G2 (15 CFU during the dry season), G3 (9.3 CFU during the rainy season), and in R7 and R12 (9.3 CFU) (Table 4). These findings are critical, as any detection of *E. coli* in water intended for human consumption, recreational activities, or irrigation of raw vegetables is deemed unacceptable according to World Health Organization (2019). Seasonality had a marginal influence on total and thermotolerant coliform levels, with slight reductions observed during the rainy season at most groundwater sampling points.

4 Discussion

This study integrated the assessment of groundwater, river, and reservoir water quality in the Iguatu, Ceará, in the Brazilian semiarid region. Physical, chemical, and microbiological parameters were evaluated during dry and rainy seasons, focusing on irrigation suitability and compliance with World Health Organization standards. Multivariate analysis and seasonal comparisons revealed substantial changes in water composition, underscoring the critical role of seasonality in driving hydrogeochemical processes. The distinction between groundwater and

surface water was reinforced by differences in salinity, ionic relationships, and potential risks for irrigation use.

The integration of site-specific historical water uses (Table 1) with the observed hydrochemical patterns provides valuable insights into the relative influence of geological and anthropogenic factors on water quality. Wells located within the sedimentary basin and primarily used for irrigating annual crops (G1, G2, G3, and G6) and fruit orchards (G4 and G10) exhibited distinct hydrochemical characteristics. While G1 and G2 showed lower salinity and hardness levels, reflecting active recharge and dilution processes within alluvial aquifers, G3 and G6 presented higher electrical conductivity and bicarbonate concentrations, which may be associated with localized agricultural practices, including irrigation return flows and nutrient inputs. In contrast, wells G8 and G9, situated within the crystalline basement and historically used for pasture and sugarcane irrigation, exhibited the highest salinity, sodium, and chloride concentrations, highlighting the dominant influence of prolonged water–rock interactions in fractured aquifers. Nevertheless, potential contributions from agricultural activities, particularly fertilizer application and livestock management, cannot be excluded.

Surface water sites further illustrate the influence of local land use. R12, located within the urban area of Iguatu and subjected to recreational activities, exhibited elevated turbidity and thermotolerant coliform counts, suggesting the influence of diffuse urban runoff or localized wastewater inputs. Conversely, the upstream (R11) and downstream (R13) stretches exhibited better microbiological conditions, indicating that contamination sources are likely spatially restricted. Despite its multiple uses, including irrigation, livestock watering, and domestic supply, the Orós Reservoir (OR5) maintained relatively stable physicochemical conditions between seasons, probably due to its large storage capacity and buffering effect. By incorporating historical land-use information into the interpretation of hydrochemical variability, water quality patterns can be understood beyond a strictly lithological perspective, facilitating the identification of priority areas for management interventions, such as riparian protection, optimized fertilizer use, and improvements in sanitation infrastructure, thereby contributing to sustainable water management in semi-arid agricultural systems.

Gibbs diagrams for groundwater and surface water in Iguatu, Ceará, Brazil

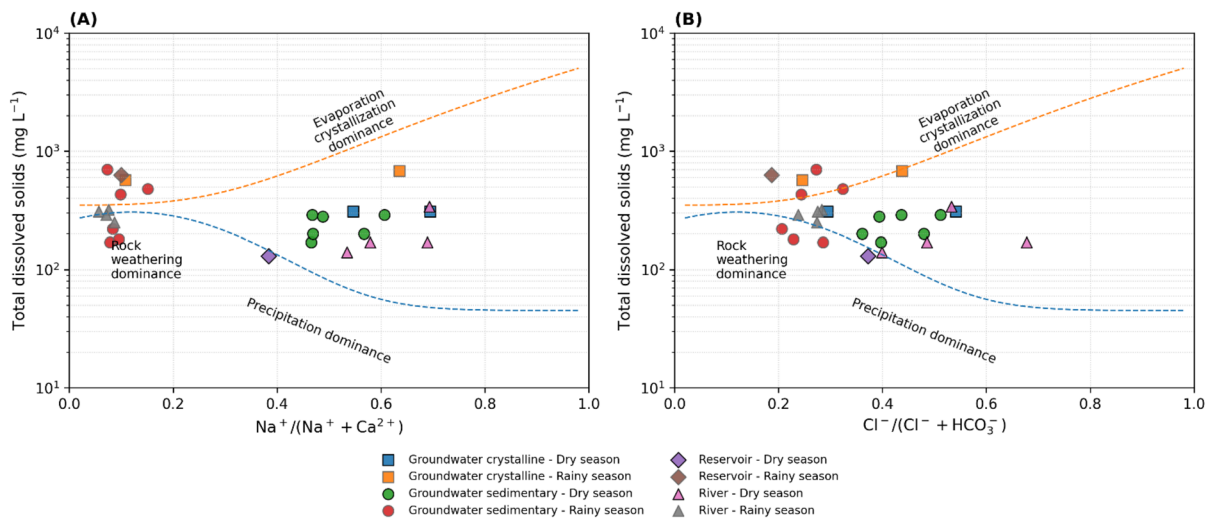


Fig. 6 Gibbs diagrams showing the relationships between TDS and (A) $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and (B) $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ for groundwater and surface water samples collected during dry and rainy seasons in the municipality of Iguatu, Ceará, Brazil

Table 3 Salinity classification of water from different water sources during dry and rainy periods based on total dissolved solids (TDS) and electrical conductivity (EC) in the municipality of Iguatu-CE, Brazil

Unit	Dry			Rainy		
	EC $\mu\text{S cm}^{-1}$	TDS mg L^{-1}	Classification	EC $\mu\text{S/cm}$	TDS mg L^{-1}	Classification
G1	400,0	260,0	Fresh water	360,0	234,0	Fresh water
G2	450,0	292,5	Fresh water	380,0	247,0	Fresh water
G3	790,0	513,5	Brackish water	700,0	455,0	Fresh water
G4	400,0	260,0	Fresh water	250,0	162,5	Fresh water
OR5	310,0	201,5	Fresh water	260,0	169,0	Fresh water
G6	870,0	565,5	Brackish water	720,0	468,0	Fresh water
R7	430,0	279,5	Fresh water	330,0	214,5	Fresh water
G8	980,0	637,0	Brackish water	940,0	611,0	Brackish water
G9	930,0	604,5	Brackish water	930,0	604,5	Brackish water
G10	740,0	481,0	Fresh water	392,0	254,8	Fresh water
R11	370,0	240,5	Fresh water	340,0	221,0	Fresh water
R12	460,0	299,0	Fresh water	350,0	227,5	Fresh water
R13	840,0	546,0	Brackish water	340,0	221,0	Fresh water

* G Groundwater, OR Oros reservoir, R Jaguaribe River, EC electrical conductivity, TDS total dissolved solids

During the dry season, higher concentrations of ions such as Mg^{2+} , Ca^{2+} , Na^+ , Cl^- , SO_4^{2-} , and HCO_3^- were observed, particularly in wells G8 and G9. Similar conditions were reported by (Rojas et al., 2020) in northeastern Brazil. This pattern is likely linked to the crystalline basement geology, composed

of igneous and metamorphic rocks (granites and gneisses) rich in primary minerals (feldspars, micas), which release Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and silica during weathering, thereby increasing salinity (Gaaloul & Eslamian, 2022; Jalali, 2007; Teramoto et al., 2023). Furthermore, due to low porosity and permeability,

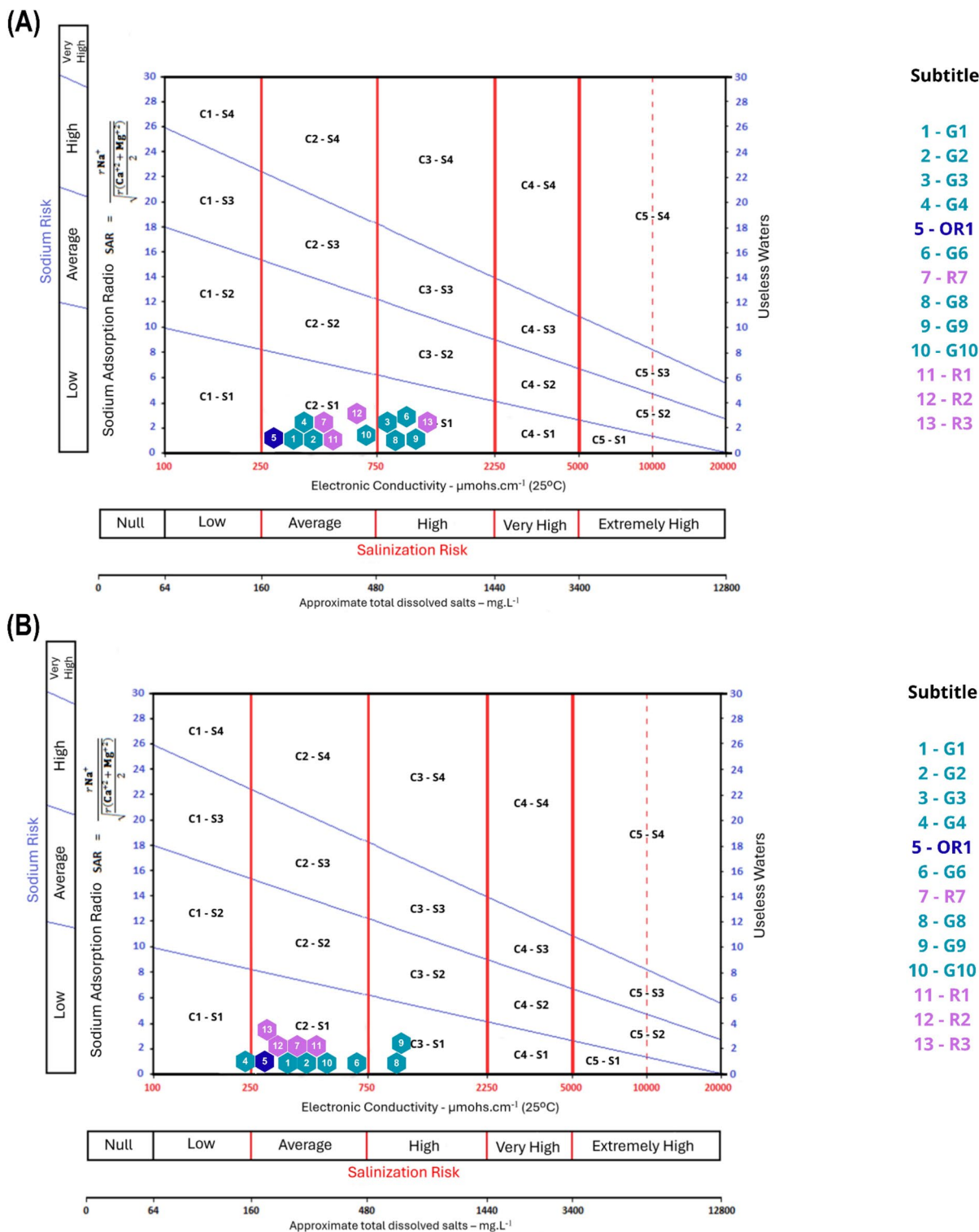


Fig. 7 Water samples collected at points distributed among groundwater wells (G1-G4, G6, G8, G9, G10—green), surface reservoir (OR5—Orós Reservoir—blue), and stretches of the

Jaguaribe River (R7, R11, R12, R13—purple), during the dry (A) and rainy (B) periods, in the municipality of Iguatu, Ceará, Brazil, plotted on U.S.S.L. Diagram

Fig. 8 Hierarchical cluster dendrogram based on Euclidean distance and Ward's linkage method for different water sources: groundwater wells (sites G1 to G4, G6, G8, G9, and G10), surface reservoir (OR5—Orós Reservoir), and stretches of the Jaguaribe River (sites R7, R11, R12, and R13), during the dry (A) and rainy (B) seasons in the municipality of Iguatu, Ceará, Brazil

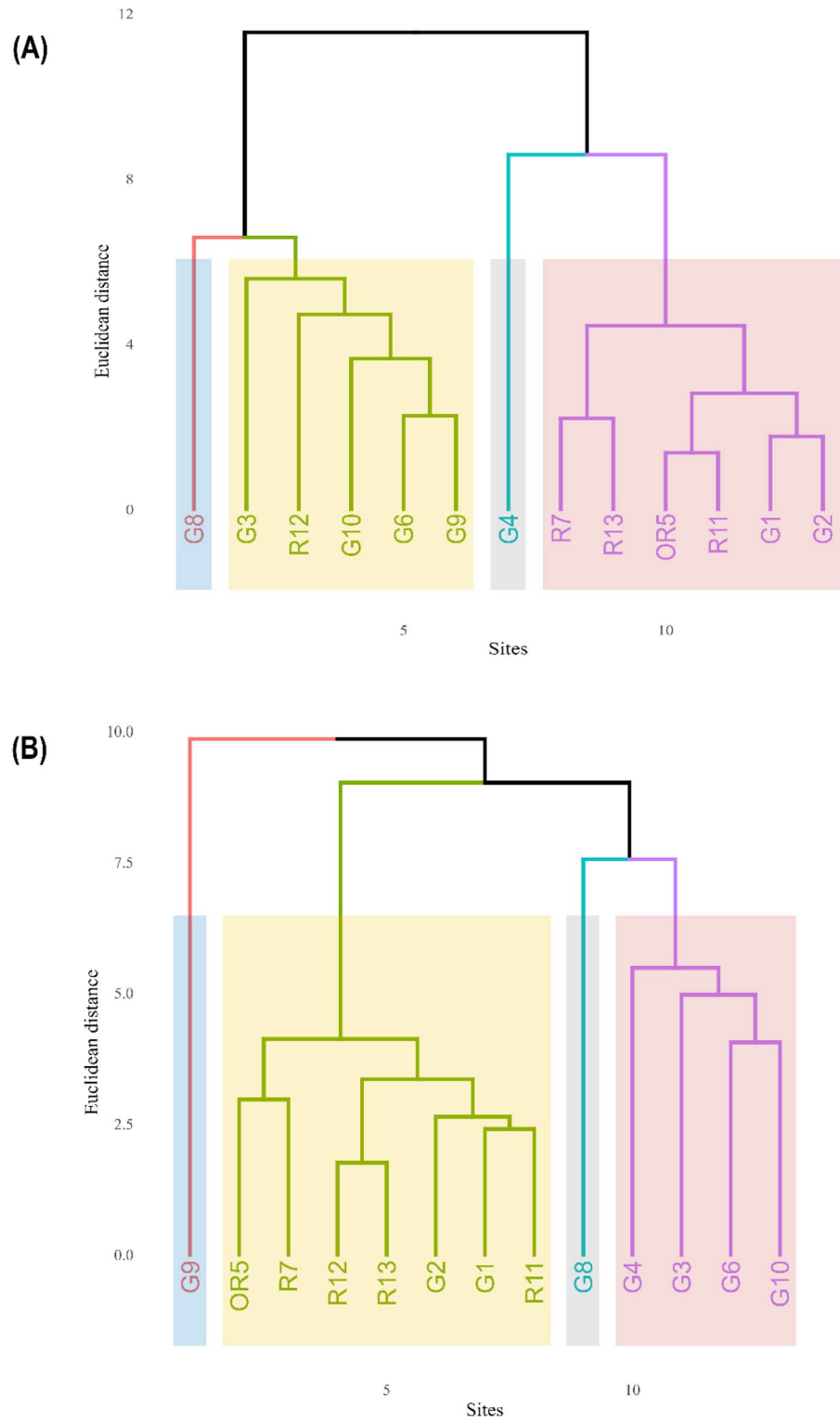


Table 4 Total and thermotolerant coliforms and *Escherichia coli* from different water sources (groundwater (G), Orós Reservoir (OR) and Jaguaribe River (R) during the dry and rainy seasons in the municipality of Iguatu, Ceará, Brazil

Sites	Total coliform (MPN mL ⁻¹)		Thermotolerant coliform (MPN mL ⁻¹)		<i>Escherichia coli</i> (CFU)	
	Dry	Rainy	Dry	Rainy	Dry	Rainy
G1	15	10	<3.0	<3.0	Absent	3.0
G2	9.3	10.3	<3.0	<3.0	15	Absent
G3	46	36	36	24	Absent	9.3
G4	24	19	<3.0	<3.0	3.0	3.0
OR5	9.3	9.9	<3.0	<3.0	Absent	Absent
G6	<3.0	<3.0	Absent	Absent	Absent	Absent
R7	46	48	36	24	Absent	9.3
G8	46	47	Absent	Absent	Absent	Absent
G9	12	15	<3.0	<3.0	Absent	3.0
G10	4.3	4.9	<3.0	<3.0	Absent	3.0
R11	<3.0	<3.0	Absent	Absent	Absent	Absent
R12	24	22	<3.0	<3.0	9.3	9.3
R13	<3.0	<3.0	6.3	4.3	Absent	0.9

*MPN most probable number, CFU colony forming unit

water flows mainly through fractures and faults, resulting in prolonged residence times and increased contact with minerals (Lachassagne et al., 2021; Wadi et al., 2022). This promotes intense dissolution and chemical alteration, leading to salt accumulation, particularly chlorides and sulfates (Egbueri et al., 2025). The low permeability of these rocks' limits salt dilution, concentrating them in fractured aquifers (Egbueri et al., 2025). Human activities such as groundwater overexploitation and improper irrigation management may further contribute to salinization (Almeida et al., 2025; Faraj et al., 2020).

Although geological factors represent the primary control on hydrochemical variability, land use and historical water use patterns also contribute to shaping water quality. Sites associated with intensive agriculture, particularly sugarcane cultivation, fruit production, and irrigated crops, generally exhibited higher concentrations of dissolved constituents, suggesting the influence of irrigation return flows, nutrient inputs, and evapoconcentration processes. Likewise, locations exposed to urban and recreational activities, such as R12, showed greater microbiological impairment, whereas the Orós Reservoir (OR5) maintained relatively stable physicochemical conditions due to its greater dilution capacity. These findings indicate that water quality patterns in the Upper Jaguaribe Basin are governed by the combined influence of geological setting and local anthropogenic pressures, reinforcing the need for integrated

monitoring strategies to support sustainable irrigation management in semi-arid regions.

Molar ratios calculated from the ionic concentrations of groundwater, the Jaguaribe River, and the Orós Reservoir (Table S2) reinforce the interpretation that water–rock interaction is a primary mechanism governing hydrogeochemical evolution in the Upper Jaguaribe Basin. The Na/Cl ratio exhibited mean values of 0.67 during the dry season and 0.27 during the rainy season, remaining consistently below unity. As reported by Jalali (2007) for arid environments, these values indicate that halite dissolution is not the sole source of salinity, suggesting the influence of ion-exchange processes, atmospheric chloride inputs, and the weathering of sodium-rich plagioclase minerals. The Ca/Mg ratio increased from 0.72 in the dry season to 1.18 in the rainy season, indicating a greater contribution from dolomite dissolution during the dry period and enhanced calcite dissolution under higher recharge conditions.

Moreover, the elevated Ca/Na ratios (0.77 in the dry season and 4.88 in the rainy season) and HCO₃/Na ratios (1.78 in the dry season and 9.80 in the rainy season), interpreted according to the framework proposed by Gaillardet et al. (1999), reveal a mixed influence of carbonate weathering associated with the Moura Formation and silicate weathering derived from the crystalline basement. The marked increase in these ratios during the rainy season suggests intensified carbonate dissolution promoted by

the infiltration of CO₂-enriched recharge waters, whereas evapoconcentration during the dry season primarily acts as a seasonal modulator, increasing ionic concentrations without substantially altering the original geochemical signatures. Collectively, these stoichiometric relationships, together with the strong Pearson correlations among Ca²⁺, Mg²⁺, and HCO₃⁻ ($r > 0.80$), demonstrate that water–rock interactions linked to regional lithological heterogeneity exert a dominant control on the chemical composition of surface and groundwater resources in the Upper Jaguaribe Basin. In contrast, the seasonal increase in salinity and SAR values during the dry season reflects the influence of evapoconcentration, consistent with observations reported for other semiarid hydrological systems (Wu et al., 2021; Gaillardet et al., 1999; Appelo & Postma, 2005).

The formation of two well-defined multivariate groups, one axis associated with sodium and chlorides, influencing SAR and EC, and another linked to carbonates and alkaline earth cations, suggests chemical dissolution processes of minerals in rock–water contact zones. In addition to dissolution from crystalline basement rocks, the presence of carbonates and alkaline earth cations is likely influenced by fracture sealing due to calcite precipitation, which reduces available permeability with depth, as observed by (Abiye et al., 2011) in crystalline rock aquifers in South Africa. Correlation analyses revealed stable hydrodynamic interactions throughout the year, such as the strong relationship between Mg²⁺, Ca²⁺, and HCO₃⁻, and between SAR and Cl⁻, indicating that sodification remains a persistent risk, particularly in waters with high Na⁺ content and low relative hardness (Ayers & Westcot, 1985; World Health Organization, 2019). These patterns suggest that even during the rainy season, groundwater quality may not fully recover from the saline concentration accumulated during the dry season, highlighting the need for more stringent water management, as demonstrated in this study.

Another critical aspect concerns the high pH values (9.6 at R7 and 8.8 at R13) recorded during the dry season, which exceed the thresholds established by FAO guidelines for irrigation water (Ayers & Westcot, 1985) and the Brazilian resolution CONAMA 357/2005 (CONAMA, 2005). These alkaline conditions were observed under peak solar radiation, suggesting a strong influence of eutrophication processes

linked to nitrogen and phosphorus enrichment. Excessive algal and macrophyte growth under such nutrient availability enhances photosynthetic activity, leading to intense CO₂ assimilation and the consequent conversion of bicarbonates (HCO₃⁻) into hydroxyl ions (OH⁻), thereby elevating pH (Beer et al., 2021; Devlin & Brodie, 2023; Dolui et al., 2021; Omokaro et al., 2025; Sitote & Gebremedhine, 2024). Elevated pH in eutrophic waters has been widely reported in tropical reservoirs, where seasonal nutrient pulses and high irradiance act synergistically to increase primary productivity and shift carbonate equilibria (Dolui et al., 2021; Omokaro et al., 2025). This dynamic not only compromises water quality for irrigation but also alters trace metal solubility and the ecological balance of aquatic systems.

The metal contents were also relevant for assessing water potability and its suitability for agricultural use. Iron (Fe) exceeded the World Health Organization limit (0.3 mg L⁻¹) in several wells during the dry season, particularly G4, G2, and G10, compromising water use due to encrustation and clogging in irrigation systems (Cabrera-Garcia et al., 2024; Pavithran et al., 2022; World Health Organization, 2019). Manganese (Mn) also exceeded acceptable limits in G3 and G4, possibly, the elevated Mn concentrations observed in wells G3 and G4, located within the sedimentary porous aquifer system, may be associated with weathering processes, the dissolution of Fe–Mn oxyhydroxides, and seasonal fluctuations in groundwater levels that promote reducing conditions and enhance metal mobilization (Appelo & Postma, 2005; World Health Organization, 2019; Latif et al., 2025). The porous nature of these aquifers may also favor longer water–sediment interaction times, contributing to increased Mn availability in solution. Although Cu, Zn, Ni, and Cr levels remained below international thresholds, their presence indicates the need for continuous monitoring, particularly in urban and agricultural areas that may be subject to anthropogenic influence.

According to the calculated indices, most sampling points showed a low risk of sodification (SAR < 10) and carbonation (RSC < 2.5) in both periods, confirming the overall suitability of the water for irrigation based on the criteria of Ayers and Westcot (1985). However, marginal RSC values observed at point R11 (1.23 in the rainy season) and SSP values exceeding or close to 50% at R12 (54.3%) and G9 (48.1%)

during the dry season warrant special attention, as they may lead to soil structure degradation over time (Kundu & Nag, 2018; Mokhtar et al., 2022). The high magnesium adsorption ratio ($MAR > 1$) at all points during the dry period, and partially in the rainy period, suggests dominance of Mg^{2+} over Ca^{2+} , which can promote colloid dispersion and compaction in clay soils, as discussed by Damiba et al. (2025). Most points also exceeded 50% in the magnesium hazard index (MH) during the dry season, especially G6, G8, OR5, and R7, implying a potential risk to the permeability of irrigated soils (Damiba et al., 2025; Kundu & Nag, 2018). Despite this, the Kelly's Index ($KI < 1$) and permeability index ($PI > 60\%$) remained at satisfactory levels, indicating a favorable ionic balance for infiltration and agricultural use, as highlighted by Ibrahim et al. (2023).

The Piper diagram indicated a strong influence of seasonality on the ionic composition of the water. In the dry period, the "mixed chloride" type predominated in groundwater, reflecting evapoconcentration, low recharge, and contact with geological materials rich in soluble salts (Aditya et al., 2024; Wu et al., 2021). During the rainy period, there was a transition to mixed bicarbonate waters, with dilution of ions and increased influence of recharge, as demonstrated by Aditya et al. (2024) and Mattos et al. (2018). It is noteworthy that the Orós Reservoir presented an intermediate composition between aquifers and the river, shifting from bicarbonate–magnesian (dry) to mixed bicarbonate (rainy), evidencing its role as a mixing and hydrological stabilization reservoir. The waters of the Jaguaribe River showed variable typologies between points and periods, ranging from mixed sodium (R12 in the dry season) to calcium bicarbonate in the rainy season, reflecting greater river renewal and dilution of ionic loads. Analyzing the Jaguaribe River, Trussu Reservoir, and groundwater also in the municipality of Iguatu-CE, Souza et al. (2017) classified the waters as chloride–calcium, bicarbonate–calcium, and bicarbonate–sodium for surface and groundwater, and chloride–sodium for groundwater. Regarding irrigation classification, their results showed that surface waters had a low risk of salinity problems, whereas groundwater showed a medium to high risk of causing salinity.

The integrated interpretation of the Gibbs and Piper diagrams confirms that water–rock interaction is the primary mechanism controlling water

chemistry in the Upper Jaguaribe Basin. In contrast, evapoconcentration acts as a secondary process of ionic enrichment. This hydrogeochemical pattern is consistent with findings reported for other arid and semi-arid environments, where mineral weathering and seasonal evaporation jointly regulate groundwater evolution (Monteiro et al., 2022; Liu et al., 2023; Razi et al., 2024). The predominance of samples within the rock-weathering dominance field highlights the strong influence of carbonate dissolution associated with the Moura Formation, as well as silicate weathering processes in the crystalline basement. In contrast, the shift of groundwater samples from crystalline aquifers, particularly G8 and G9 during the dry season, toward the evaporation–crystallization domain suggests that longer residence times in fractured aquifers, combined with reduced recharge, favor progressive solute accumulation.

Surface waters from the Jaguaribe River and the Orós Reservoir occupy intermediate positions between the precipitation and rock-weathering domains, reflecting the combined influence of seasonal recharge and lithological controls. The seasonal migration of samples toward higher salinity conditions during the dry period, followed by their return to the rock-weathering domain during the rainy season, indicates that hydroclimatic variability primarily modulates ionic concentrations without substantially altering the dominant hydrogeochemical signatures inherited from mineral dissolution processes. Overall, the combined interpretation of the Piper and Gibbs diagrams demonstrates that hydrochemical evolution in the Upper Jaguaribe Basin is fundamentally governed by water–rock interactions, whereas evapoconcentration and meteoric recharge act as secondary drivers regulating salinity dynamics at the seasonal timescale.

According to the parameters of CONAMA Resolution No. 357/2005, during the dry season, five of the eight wells (G3, G6, G8, G9, and G10) were classified as brackish water—a typical effect of intense evaporation and mineralization in shallow crystalline aquifers (Teramoto et al., 2023). During the rainy season, six wells showed significant dilution, leading to reclassification as freshwater, except for G8 and G9, whose salinity remained high. The Orós Reservoir demonstrated excellent hydrological stability, which can likely be attributed to its large storage volume, continuous renewal from river inflows, and the intrinsic

dilution capacity of the water body. The waters of the Jaguaribe River also exhibited good quality in both seasons, with a predominance of freshwater classification, except at point R13 during the dry season (classified as brackish water). This localized increase may be related to concentrated discharges, saline surface runoff, or longer local residence times, particularly in stretches with reduced flow velocity or impoundment.

The assessment using the U.S. Salinity Laboratory diagram revealed that, although many points fell into the C3-S1 class (high salinity and low sodicity hazard), after the rainy season there was a migration to C2-S1 or even C1-S1, highlighting the crucial role of rainfall in reducing salinity and improving water suitability for irrigation (Liberoff & Poca, 2023; McGeorge, 1954; Perri et al., 2022). Even so, G8 and G9 remained in the C3-S1 class, requiring restrictions on agricultural use or the adoption of specific management practices (e.g., gypsum application or cultivation of salt-tolerant crops).

The hierarchical cluster analysis (HCA) confirmed the influence of water source origin and seasonality on the distribution of chemical patterns. In the dry season, G8 stood out as an isolated point due to its high EC, Na^+ , and Cl^- levels, representing a typical case of localized salinization. In the rainy season, G8 and G9 were isolated, the latter associated with higher hardness and alkaline earth cations. Distinct clusters grouped wells with similar composition (G3, G6, G10) and clustered surface waters (river and reservoir) with lower-salinity wells (G1 and G2), indicating strong compositional similarity. During the rainy season, greater homogeneity between groundwater and surface water was observed, with the grouping of points (OR5, R11, R13, G1, G2) into a single cluster, reinforcing the role of recharge and mixing of water masses in attenuating chemical contrasts (Carvalho et al., 2021; Perri et al., 2022; Tinôco et al., 2018).

The microbiological analysis revealed widespread contamination by total coliforms in all water samples, both surface and groundwater, during the dry and rainy seasons. Although values varied between sampling points, the consistent presence of these microorganisms is alarming, particularly in wells G3, G4, and G8, and in river waters at R7 and R12, where total coliform levels reached up to 46 MPN mL^{-1} , indicating a potential risk to human health. Shallow groundwater sources are more susceptible to contamination,

as the short vertical distance facilitates the transport of pathogenic microorganisms. The effectiveness of reducing these contaminants along the subsurface flow depends on their retention and microbiological inactivation during the transit (van Geen et al., 2011).

According to the World Health Organization (2019), total coliforms in any type of water intended for human consumption, irrigation of crops eaten raw, or recreational use represent non-compliance with potability and microbiological safety standards, requiring immediate corrective measures. Interpreting microbiological results within the context of historical water uses (Table 1) reveals distinct contamination patterns associated with local anthropogenic activities. Wells G3 and G4, located within the sedimentary basin and intensively used for irrigating annual crops and banana plantations, exhibited the highest total coliform counts (up to 46 MPN mL^{-1}), suggesting diffuse microbial inputs derived from agricultural practices, irrigation return flows, organic amendments, and the possible influence of inadequate rural sanitation systems, including septic tank leakage. In contrast, G6, historically used for pasture irrigation and livestock watering, maintained negligible coliform levels (<3.0 MPN mL^{-1}), indicating that appropriately managed water sources can preserve satisfactory microbiological quality when protected from direct surface contamination.

Attention should be given to G2, which serves as an alternative domestic water supply during drought periods and presented detectable *Escherichia coli* levels (15 CFU), highlighting potential public health concerns related to untreated groundwater use. Among surface water sites, R12, located within the urban area of municipality of Iguatu and subjected to recreational activities, consistently exhibited elevated coliform and *E. coli* counts (9.3 CFU), suggesting the influence of diffuse urban runoff and localized wastewater inputs. Conversely, the upstream (R11) and downstream (R13) stretches, which are less exposed to urban pressures, showed comparatively lower microbial loads. Despite its multiple uses, including irrigation, livestock watering, and domestic supply, the Orós Reservoir (OR5) maintained relatively low contamination levels, likely due to its large storage capacity and greater dilution potential. Collectively, these findings indicate that microbiological impairment is spatially heterogeneous and reflects the combined influence of land use, water use history, and

local anthropogenic pressures, emphasizing the need for targeted management measures to safeguard water quality and strengthen water security in semi-arid agricultural regions.

The detection of *Escherichia coli* (*E. coli*) at sites such as G2 (15 CFU), G3 (9.3 CFU), and R12 (9.3 CFU) is particularly critical, as this bacterium is a direct marker of microbiological risk, associated with enteric pathogens such as *Salmonella* spp., *Shigella* spp., and pathogenic strains of *E. coli* (Dzodzomenyo et al., 2022). According to World Health Organization (2019), any detectable concentration of *E. coli* in 100 mL of sample renders the water unsuitable for human consumption and also unfit for irrigating crops eaten raw, due to the high risk of transmitting diseases such as colibacillosis, hepatitis A, giardiasis, diarrhea, cholera, typhoid fever, salmonellosis, and poliomyelitis (Bagordo et al., 2024; Silva et al., 2024; World Health Organization, 2019).

Seasonality had a differentiated impact on microorganisms. A slight reduction in total coliforms was observed during the rainy season, possibly due to dilution from higher precipitation volumes. However, the presence of *E. coli* worsened at some points during this same period, such as in wells G3 and G9 and river points R7 and R12, suggesting that surface runoff and recharge of contaminated aquifers may have transported fecal matter to water sources (Tagar et al., 2025) (Dzodzomenyo et al., 2022; Tagar et al., 2025). From a practical standpoint, these results indicate a high risk to public health, especially in areas where these waters are used for irrigating vegetables eaten raw, such as lettuce, coriander, and scallions, and where recreational activities take place, as in the Jaguaribe River. Consumption of food irrigated with contaminated water can lead to outbreaks of waterborne diseases, with direct impacts on food safety, the healthcare system, and the local economy.

5 Conclusions

This study demonstrates that hydrogeochemical evolution in the Upper Jaguaribe Basin is primarily governed by water–rock interactions, while seasonal evapoconcentration acts as a secondary mechanism enhancing salinity under dry conditions. The integrated application of hydrochemical indices, multivariate statistics, and geochemical classification tools

revealed pronounced spatial and seasonal heterogeneity, particularly between sedimentary and crystalline aquifers, highlighting the strong influence of lithological controls on water quality and irrigation suitability. The occurrence of microbial contamination in selected groundwater and surface water sources further emphasizes the vulnerability of water resources to local anthropogenic pressures, including agricultural activities, inadequate sanitation infrastructure, and diffuse urban inputs.

From a practical perspective, these findings provide a robust scientific basis for prioritizing monitoring programs, improving rural sanitation, and implementing adaptive management strategies aimed at mitigating salinization risks, preserving soil quality, and protecting public health. The identification of hydrochemical and microbiological hotspots offers valuable support for decision-making processes related to irrigation planning, sustainable water allocation, and climate adaptation policies in semi-arid regions. Ultimately, this integrated framework contributes to strengthening water security and agricultural resilience under increasing climatic variability and anthropogenic pressures, supporting the implementation of Sustainable Development Goals 6, 12, 13, and 15.

Given the environmental and climatic vulnerability of the semi-arid region, this study recommends strengthening continuous water monitoring, expanding basic sanitation, and adopting social technologies that promote sustainable water use. Furthermore, by identifying hotspots of fecal coliform contamination and the risks associated with irrigating with low-quality water, this study reinforces the importance of SDG 3 (Good Health and Well-being), particularly in vulnerable rural communities. Therefore, the findings of this research are essential for guiding technical, environmental, and social decisions, fostering water security and agricultural resilience in contexts of scarcity and anthropogenic pressure in the Brazilian semi-arid region.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing interests The authors declare that they have no conflict of interest.

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